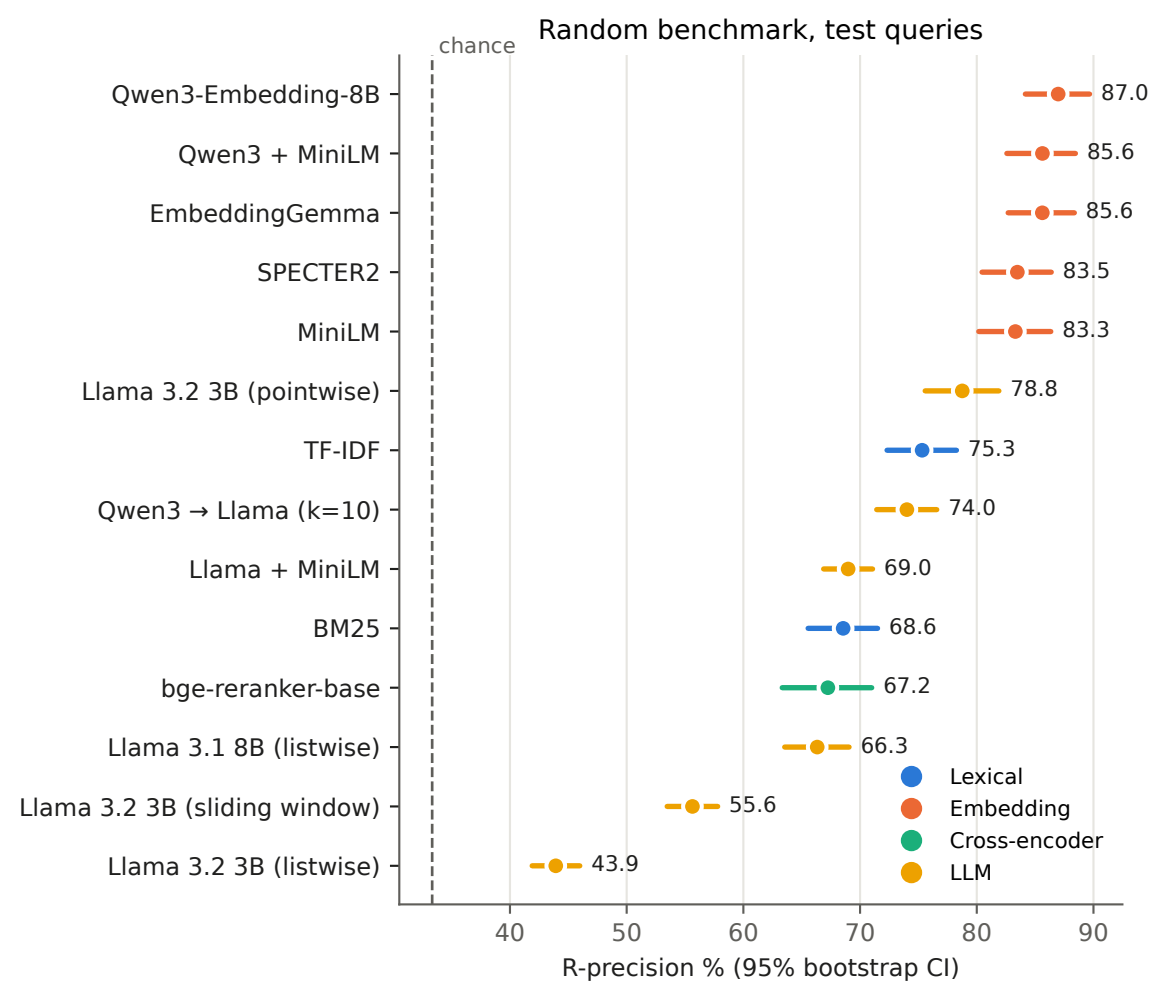
Random benchmark, test queries
chance
Qwen3-Embedding-8B
Qwen3 + MiniLM
EmbeddingGemma
SPECTER2
MiniLM
Llama 3.2 3B (pointwise)
TF-IDF
Qwen3 → Llama (k=10)
Llama + MiniLM
BM25
bge-reranker-base
Llama 3.1 8B (listwise)
Llama 3.2 3B (sliding window)
Llama 3.2 3B (listwise)
87.0
85.6
85.6
83.5
83.3
78.8
75.3
74.0
69.0
68.6
67.2
66.3
55.6
43.9
Lexical
Embedding
Cross-encoder
LLM
40
50
60
70
80
90
R-precision % (95% bootstrap CI)